

Systems Analysis and Design

5th Edition

Chapter 11. Data Storage Design

Alan Dennis, Barbara Haley Wixom, and Roberta Roth

Chapter 11 Outline

- Data storage formats.
 - Files.
 - Databases.
- Moving from logical to physical data models.
- Optimizing data storage.

INTRODUCTION

- The **data storage function** is concerned with how data is stored and handled by programs that run the system.
- Data storage design is to
 - select the data storage format;
 - convert the logical data model created during analysis into a ***physical data model*** to reflect the implementation decision;
 - ensure that DFDs and ERDs balance; and
 - design the selected data storage format to optimize its processing efficiency.

Data Storage Formats

- Two main types of data storage formats:
 - Files: electronic lists of data, optimized to perform a particular transaction
 - Database: a collection of groupings of information that are related to each other in some way
- Database Management System (DBMS): software that creates and manipulates the databases

Files

- A *data file* contains an electronic list of information that is formatted for a particular transaction.
- Typically, files are organized sequentially.
- Records can be associated with other records by *pointers*.
- Sometimes files are called *linked Lists* because of the way the records are linked together using pointers.

Types of Files

- ***Master files*** – store core information that is important to the application
- ***Look-up files*** – contain static values
- ***Transaction files*** – store information that can be used to update a master file
- ***Audit files*** – record “before” and “after” images of data as the data is altered
- ***History files*** (or archive files) – store past transactions

File based DB: appointment record

Appointment Date	Appointment Time	Duration	Reason	Patient ID	First Name	Last Name	Phone Number	Doctor ID	Doctor Last Name
11/23/2012	2:30	.25 hour	Flu	758843	Patrick	Dennis	548-9456	V524625587	Vroman
11/23/2012	2:30	1 hour	Physical	136136	Adelaide	Kin	548-7887	T445756225	Tantalo
11/23/2012	2:45	.25 hour	Shot	544822	Chris	Pullig	525-5464	V524625587	Vroman
11/23/2012	3:00	1 hour	Physical	345344	Felicia	Marston	548-9333	B544742245	Brousseau
11/23/2012	3:00	.5 hour	Migraine	236454	Thomas	Bateman	667-8955	V524625587	Vroman
11/23/2012	3:30	.5 hour	Muscular	887777	Ryan	Nelson	525-4772	V524625587	Vroman
11/23/2012	3:30	.25 hour	Shot	966233	Peter	Todd	667-2325	T445756225	Tantalo
11/23/2012	3:45	.75 hour	Muscular	951657	Mike	Morris	663-8944	T445756225	Tantalo
11/23/2012	4:00	1 hour	Physical	223238	Ellen	Whitener	525-8874	B544742245	Brousseau
11/23/2012	4:00	.5 hour	Flu	365548	Jerry	Starsia	548-9887	V524625587	Vroman
11/23/2012	4:30	1 hour	Minor surg	398633	Susan	Perry	525-6632	V524625587	Vroman
11/23/2012	4:30	.5 hour	Migraine	222577	Elizabeth	Gray	667-8400	T445756225	Tantalo
11/24/2012	8:30	.25 hour	Shot	858756	Elias	Awad	663-6364	T445756225	Tantalo
11/24/2012	8:30	1 hour	Minor surg	232158	Andy	Ruppel	525-9888	V524625587	Vroman
11/24/2012	8:30	.25 hour	Flu	244875	Rick	Grenci	548-2114	B544742245	Brousseau
11/24/2012	8:45	.5 hour	Muscular	655683	Eric	Meier	667-0254	T445756225	Tantalo
11/24/2012	8:45	1 hour	Physical	447521	Jane	Pace	548-0025	B544742245	Brousseau
11/24/2012	9:30	.5 hour	Flu	554263	Trey	Maxham	663-8547	V524625587	Vroman

FIGURE 11-1
Appointment File

Example of database:

Appointment Date	Appointment Time	Duration	Reason	Patient ID	Doctor ID
11/23/2012	2:30	.5 hour	Flu	758843	V524625587
11/23/2012	2:30	1 hour	Physical	136136	T445756225
11/23/2012	2:45	.25 hour	Shot	544822	V524625587
11/23/2012	3:00	1 hour	Physical	345344	B544742245
11/23/2012	3:00	.5 hour	Migraine	236454	V524625587
11/23/2012	3:30	.5 hour	Muscular	887777	V524625587
11/23/2012	3:30	.25 hour	Shot	966233	T445756225
11/23/2012	3:45	.75 hour	Muscular	951657	T445756225
11/23/2012	4:00	1 hour	Physical	223238	B544742245
11/23/2012	4:00	.5 hour	Flu	365548	V524625587
11/23/2012	4:30	1 hour	Minor surg	398633	V524625587
11/23/2012	4:30	.5 hour	Migraine	222577	T445756225
11/24/2012	8:30	.25 hour	Shot	858756	T445756225
11/24/2012	8:30	1 hour	Minor surg	232158	V524625587
11/24/2012	8:30	.25 hour	Flu	244875	B544742245
11/24/2012	8:45	.5 hour	Muscular	655683	T445756225
11/24/2012	8:45	1 hour	Physical	447521	B544742245
11/24/2012	9:30	.5 hour	Flu	554263	V524625587

Tables related by patient ID

Tables related by doctor ID

Patient ID	First Name	Last Name	Phone Number
136136	Adelaide	Kin	548-7887
222577	Elizabeth	Gray	667-8400
223238	Ellen	Whitener	525-8874
232158	Andy	Ruppel	525-9888
236454	Thomas	Bateman	667-8955
244875	Rick	Grenci	548-2114
345344	Felicia	Marston	548-9333
365548	Jerry	Starsia	548-9887
398633	Susan	Perry	525-6632
447521	Jane	Pace	548-0025
544822	Chris	Pullig	525-5464
554263	Trey	Maxham	663-8547
655683	Eric	Meier	667-0254
758843	Patrick	Dennis	548-9456
858756	Elias	Awad	663-6364
887777	Ryan	Nelson	525-4772
951657	Mike	Morris	663-8944
966233	Peter	Todd	667-2325

Doctor ID	Last Name
B544742245	Brousseau
T445756225	Tantalo
V524625587	Vroman

FIGURE 11-2
Appointment Database

Databases

- There are many types of databases:
 - Relational database
 - Object database
 - Multidimensional database
 - NoSQL database

Relational Databases

- **The relational database** is the most popular kind of database for application development today.
- A relational database is based on collections of **tables**, each of which has a **primary key**.
- The tables are related to each other by the placement of the primary key from one table into the related table as a **foreign key**.
- Most relational database management systems (RDBMS) support **referential integrity**, or the idea of ensuring that values linking the tables together are valid and correctly synchronized.
- **Structured Query Language (SQL)** is the standard language for accessing the data in the tables.

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Relational database example

Cust ID is the primary key of Customer

Customer

Cust ID	Last Name	First Name
4254	Baily	Ryan
9500	Chin	April
1556	Fracken	Chris
2487	Hancock	Bill
2243	Harris	Linda
1035	Black	John
1123	Williams	Mary
9501	Kaplan	Bruce
4453	Min	Julie
9505	Marvin	Sandra
2282	Lau	Mark
5927	Lee	Diane
2241	Jones	Chris
2242	DeBerry	Ann
2274	Goodlin	Dan
9507	Nelson	Dave
2264	White	Anthony

Order

Order Number	Date	Cust ID	Amount	Payment Type
234	11/23/11	4254	\$30.00	MC
235	11/23/11	9500	\$20.00	VISA
236	11/23/11	1556	\$20.00	VISA
237	11/23/11	2487	\$60.00	AMEX
238	11/23/11	2243	\$50.00	MC
239	11/23/11	1035	\$50.00	AMEX
240	11/23/11	1556	\$20.00	VISA
241	11/23/11	1123	\$40.00	MC
242	11/24/11	9501	\$30.00	VISA
243	11/24/11	4453	\$30.00	VISA
244	11/24/11	9505	\$20.00	VISA
245	11/24/11	2282	\$20.00	AMEX
246	11/24/11	5927	\$60.00	MC
247	11/24/11	2241	\$50.00	VISA
248	11/24/11	4254	\$50.00	AMEX
249	11/24/11	2242	\$50.00	AMEX
250	11/24/11	2274	\$20.00	MC
251	11/24/11	9507	\$10.00	MC
252	11/24/11	2487	\$60.00	MC
253	11/24/11	2264	\$40.00	AMEX

Cust ID is a foreign key in Order.

Payment Type is a foreign key in Order.

Referential integrity:

- All Payment Type values in Order must exist first in the Payment Type table.
- All Cust ID values in Order must exist first in the Customer table.

Payment Type

Payment Type	Description
MC	Mastercard
VISA	VISA
AMEX	American Express

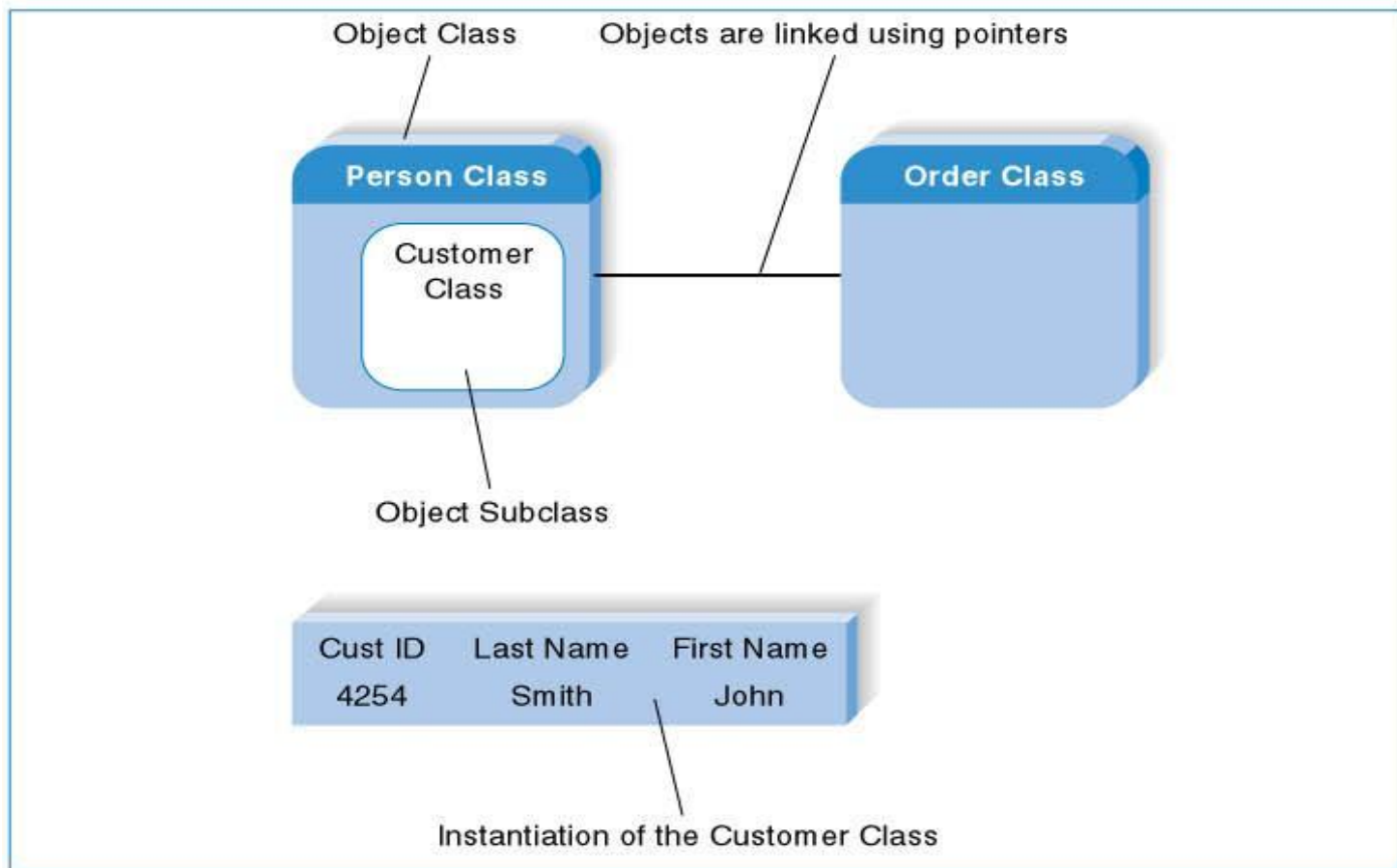
Payment Type is the primary key of the Payment Type table.

Object Databases

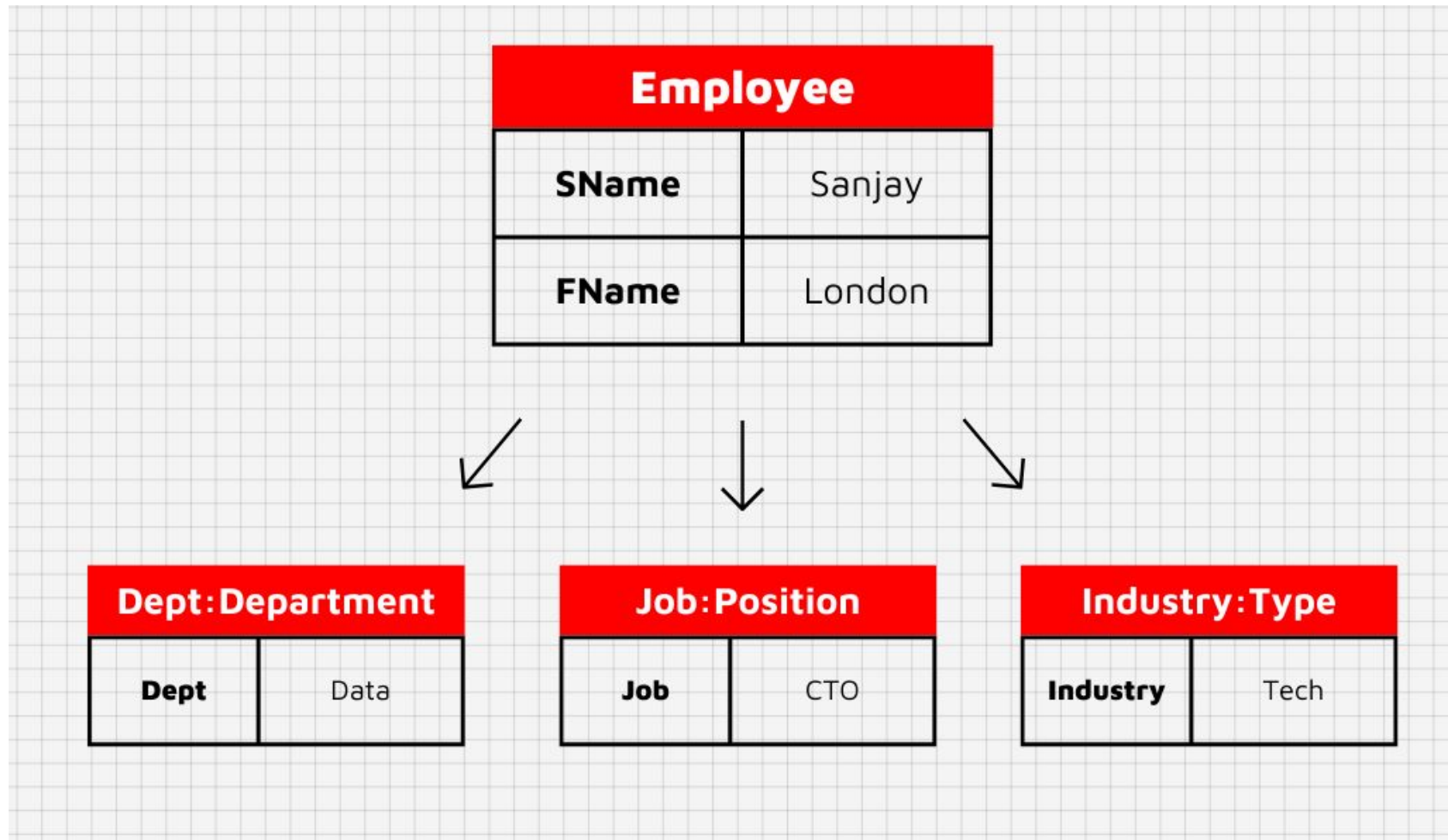
- The *object database*, or object-oriented database, is based on the premise of object orientation that all things should be treated as *objects* that have both data (attributes) and processes (behaviors).
- Changes to one object have no effect on other objects because the attributes and behaviors self-contained, or encapsulated, within each one.
- This *encapsulation* allows objects to be reused.

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■ Object Database Example

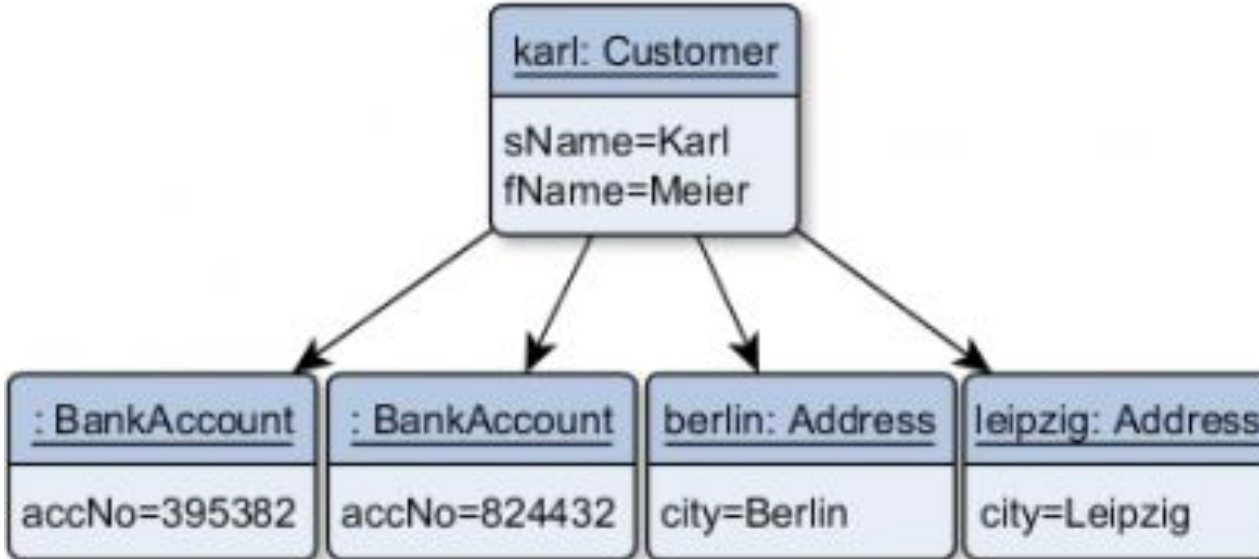


■ Object Database Example

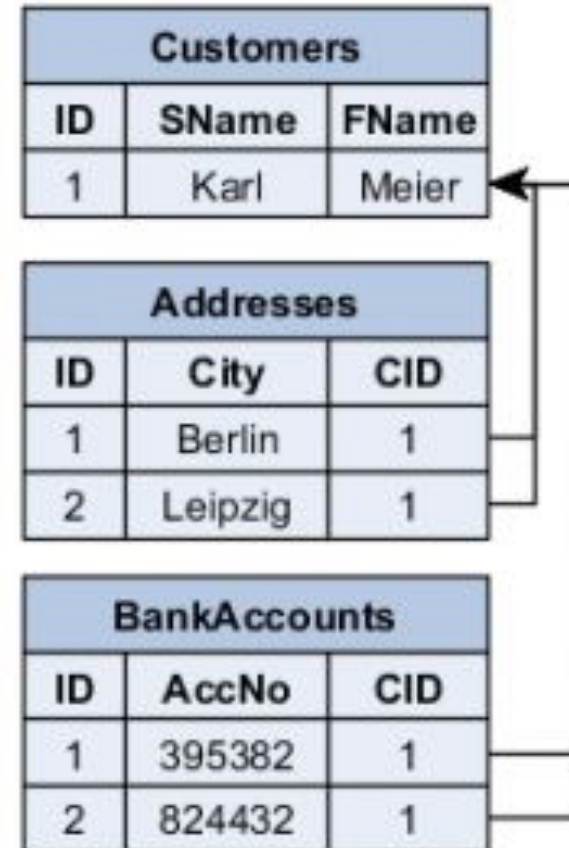


■ Object Database Example

Object-Oriented Data Model



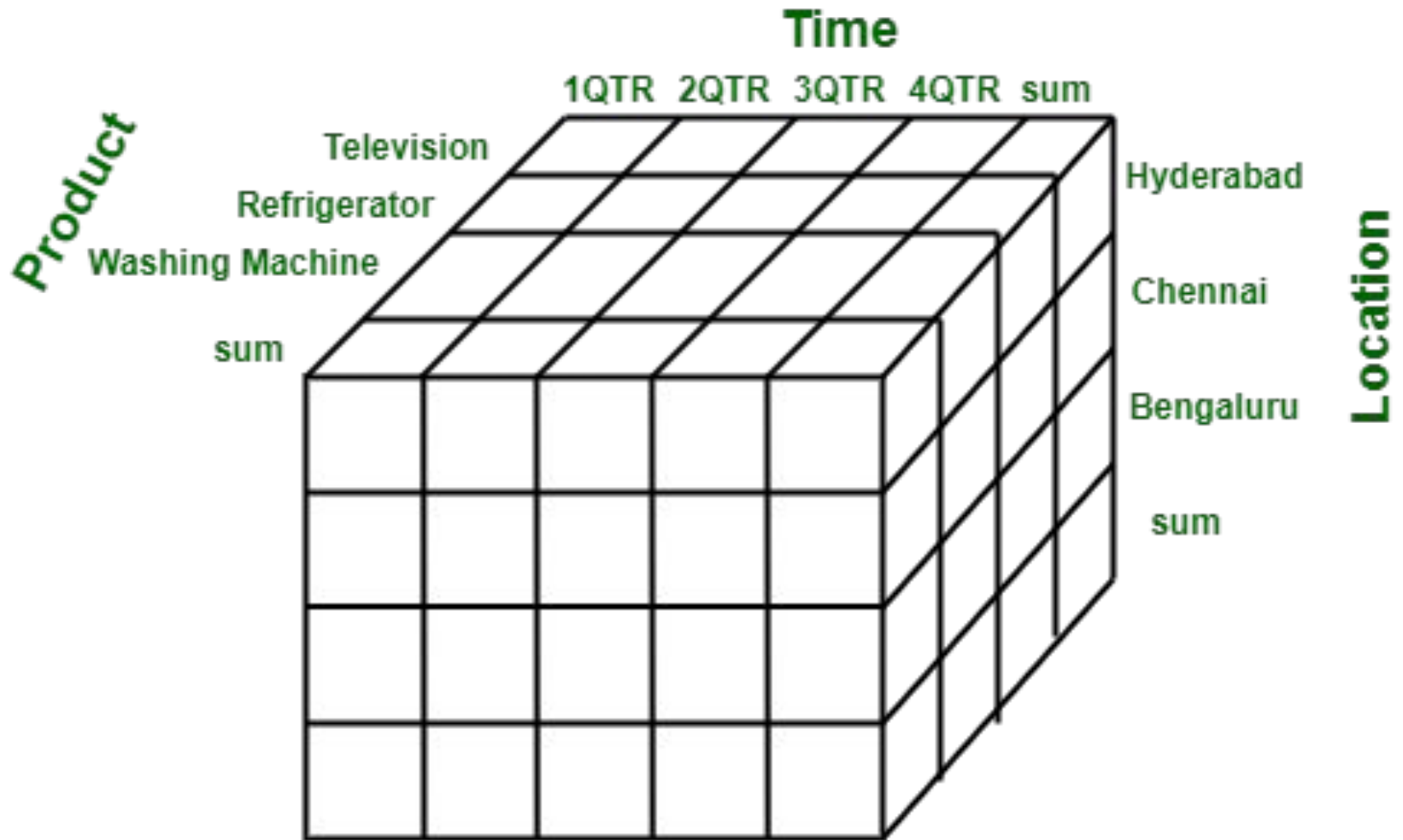
Relational Data Model



Multidimensional Databases

- A *multidimensional database* is a type of relational database that is used extensively in data warehousing.
- *Data warehousing* is the practice of taking and storing data in a data warehouse (i.e., a large database) that supports *decision support systems (DSS)*.
- *Data marts* are smaller databases based on data warehouse data, and support DSS for specific departments or functional areas of the organization.

Multidimensional Databases



NoSQL Databases

- Newest database approach; not based on the relational model or SQL
- Rapid processing on replicated database servers in the cloud
- Various types include:
 - **Document-oriented databases:** manage collection of documents of varying forms and structures (example: Mongo DB)
 - **Wide column databases:** store data in records holding very large numbers of dynamic columns (potentially billions of columns). Example: Bigtable, Cassandra, Dynamo
 - **Graph databases:** a collection of nodes and edges using graph theory to store, map, and query relationships

Selecting a Storage Format

	Files	Legacy databases	Relational databases	Object databases	Multidimensional databases	NoSQL databases
Major strengths	Files can be designed for fast performance; good for short-term data storage.	Very mature products	Leader in the database market; can handle fast updating and querying needs	Able to handle complex data	Configured to answer business intelligence questions quickly	Designed for huge, varied data sets
Major weaknesses	Redundant data; data must be updated using programs.	Not able to store data as efficiently; limited future	Cannot handle complex data	Limited market acceptance; skills are hard to find.	Highly specialized use; skills are hard to find	New in the market, highly specialized use; skills are hard to find
Data types supported	Simple	Not recommended for new systems	Simple	Complex (e.g., video, audio, images)	Aggregated	Mixed data sets with structured and unstructured components
Application system types supported	Transaction processing	Not recommended for new systems	Transaction processing and decision making	Transaction processing	Business intelligence	Business intelligence; finding patterns and relationships in mixed data
Existing data formats	Organization dependent	Organization dependent	Organization dependent	Organization dependent	Organization dependent	Organization dependent
Future needs	Limited future prospects	Poor future prospects	Good future prospects	Uncertain future prospects	Uncertain future prospects	New, uncertain future prospects

MOVING FROM LOGICAL TO PHYSICAL

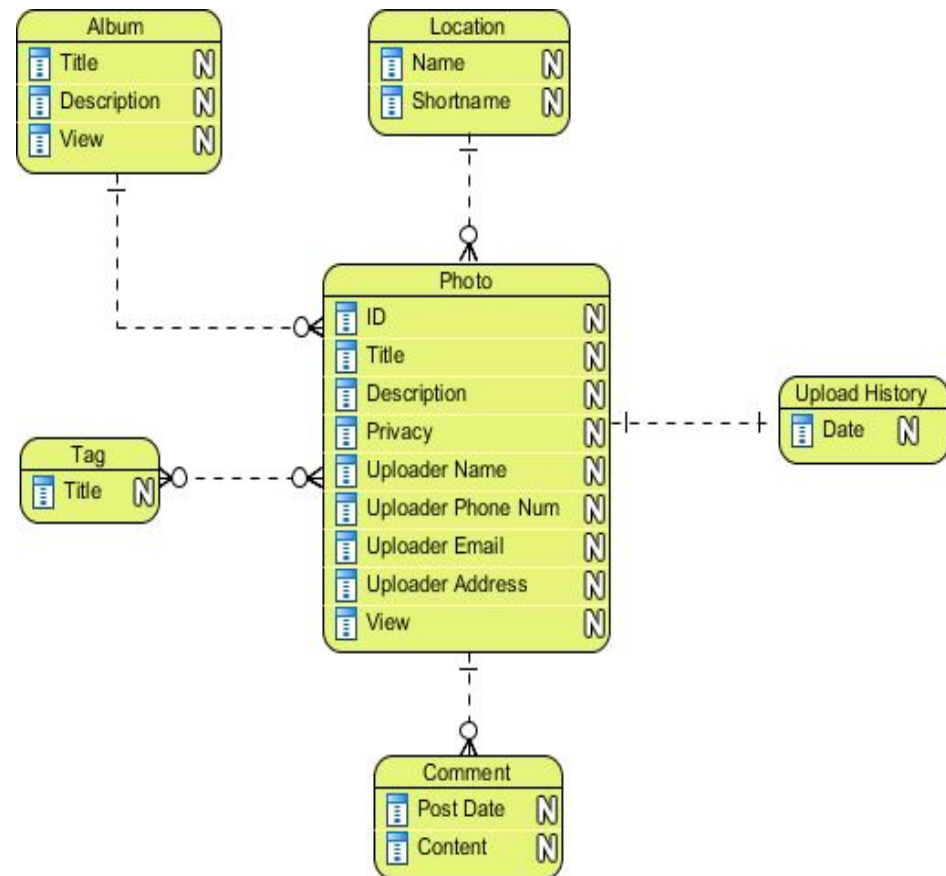
Step	Explanation
Change entities to tables or files.	Beginning with the logical entity relationship diagram, change the entities to tables or files and update the metadata.
Change attributes to fields.	Convert the attributes to fields and update the metadata.
Add primary keys.	Assign primary keys to all entities.
Add foreign keys.	Add foreign keys to represent the relationships among entities.
Add system-related components.	Add system-related tables and fields.

Conceptual, Logical and Physical

- Conceptual, logical and physical model or ERD are three different ways of modeling data in a domain.
- Business analyst uses conceptual and logical model for modeling the data required and produced by system from a business angle
- while database designer refines the early design to produce the physical model for presenting physical database structure ready for database construction.

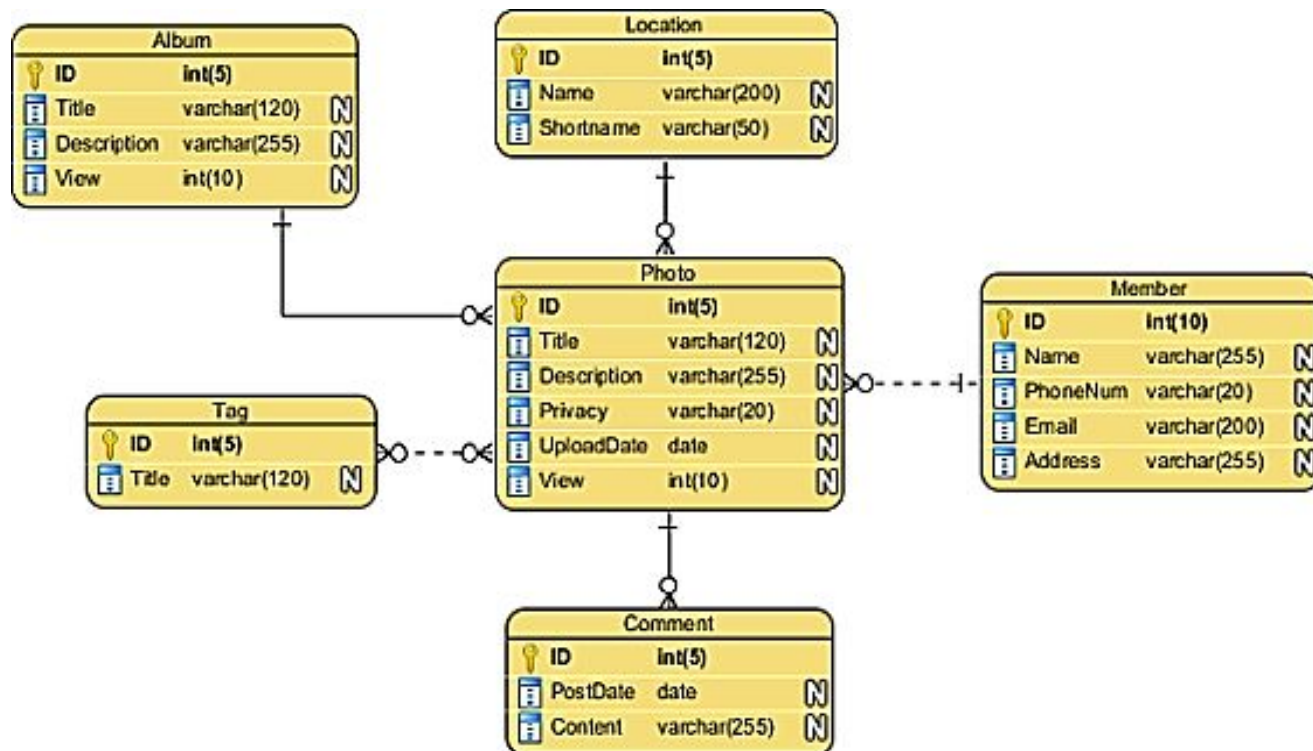
Conceptual Model

- Conceptual ERD models information gathered from business requirements.
- Entities and relationships modeled in such ERD are defined around the business's need.
- The need of satisfying the database design is not considered yet.



Logical Model

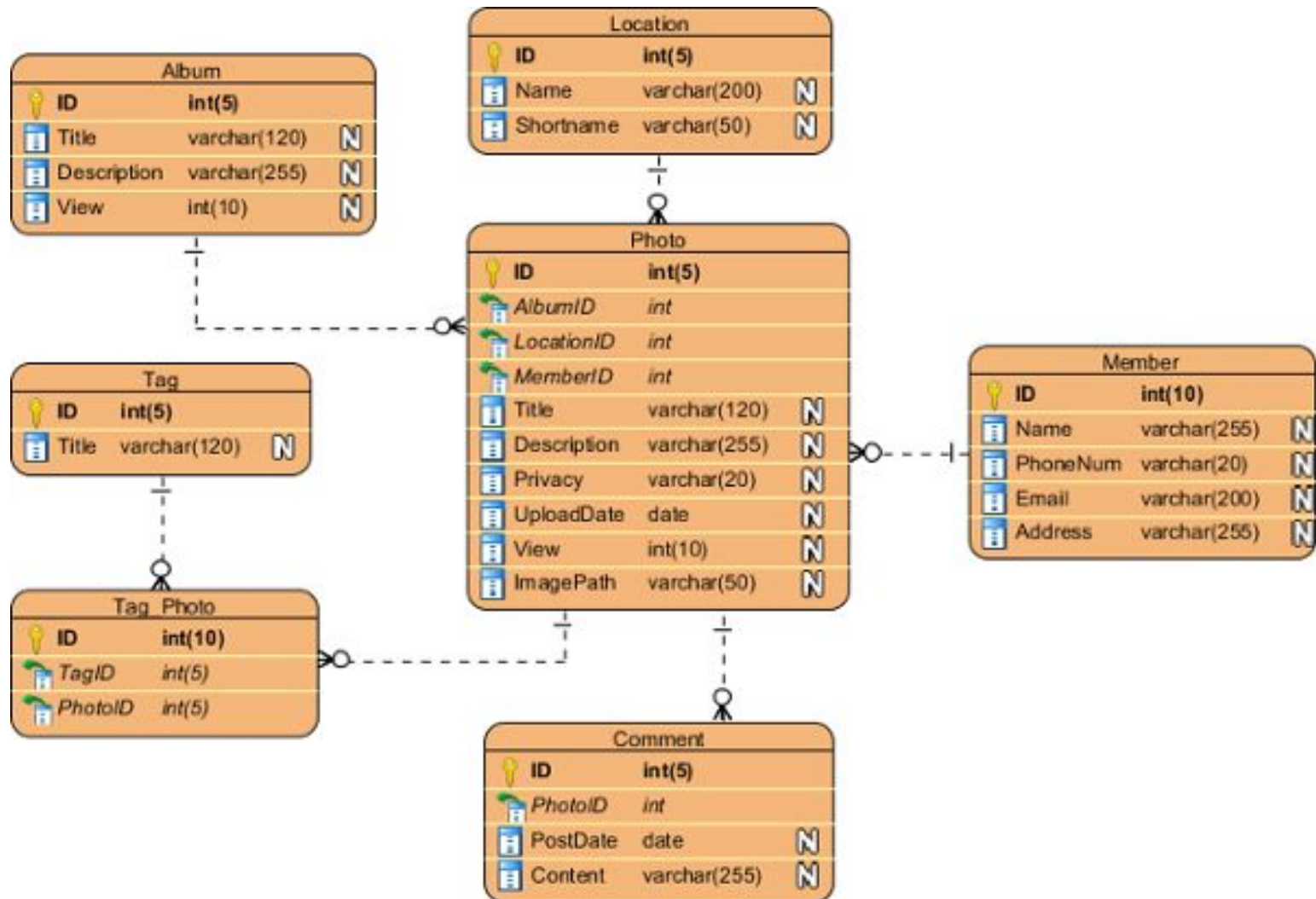
- Logical ERD also models information gathered from business requirements. It is more complex than conceptual model in that column types are set. Note that the setting of column types is optional and if you do that, you should be doing that to aid business analysis. It has nothing to do with database creation yet.



Physical Model

- Actual design blueprint of a relational database.
- It represents how data should be structured and related in a specific DBMS so it is important to consider the convention and restriction of the DBMS you use when you are designing a physical ERD.
- This means that an accurate use of data type is needed for entity columns and the use of reserved words has to be avoided in naming entities and columns.
- Besides, database designers may also add primary keys, foreign keys and constraints to the design.

Physical Model



Metadata

The screenshot shows the 'Column Editor' dialog box for the 'CUSTOMER' table. The 'Column' list on the left includes 'cust_id' (marked with a key icon), 'cust_fname', 'cust_lname', 'cust_country', 'cust_address', 'cust_city', 'cust_state', 'cust_zip', and 'cust_email'. The 'General' tab is selected, showing the 'Attribute: cust_id' and 'ORACLE Datatype' as 'CHAR(10)'. The 'Null Option' is set to 'NOT NULL'. The 'Valid' and 'Default' fields are empty. The 'Average Width' and 'Percent NULL' fields are also empty. The 'OK' and 'Cancel' buttons are at the bottom right.

Naming conventions for fields: 4 digits of table name followed by the field name.

Notice that this will be implemented in Oracle.

No null, or blank, values will be accepted into the *cust_id* field.

The key signifies that *cust_id* is a primary key.

CHAR stands for "character" data type; the 10 stands for the number of characters.

The analyst can specify a default value that appears for this field.

The analyst can develop a validation rule to be applied to this field.

OPTIMIZING DATA STORAGE

- The data storage format is now optimized for processing **efficiency**.
- There are two primary dimensions in which to optimize a relational database: for **storage efficiency** and for **speed of access**.

Optimizing Storage Efficiency

- The most efficient tables in a relational database in terms of storage space have **no redundant data** and very few null values.
- **Normalization** is the best way to optimize data storage for efficiency.

CUSTOMER ORDER

Order Number

Date
Cust ID
Last Name
First Name
State
Amount
Tax Rate
Product 1
Product Description 1
Product 2
Product Description 2
Product 3
Product Description 3

Redundant data

Null cells

Order Number	Date	Cust ID	Last Name	First Name	State	Amount	Tax Rate	Product	Product Desc	Product	Product Desc	Product	Product Desc
239	11/23/09	1135	Black	John	MD	\$50.00	0.05	555	Cheese Tray				
260	11/24/09	1135	Black	John	MD	\$40.00	0.05	444	Wine Gift Pack				
273	11/27/09	1135	Black	John	MD	\$20.00	0.05	222	Bottle Opener				
241	11/23/09	1123	Williams	Mary	CA	\$40.00	0.08	444	Wine Gift Pack				
262	11/24/09	1123	Williams	Mary	CA	\$20.00	0.08	222	Bottle Opener				
267	11/27/09	1123	Williams	Mary	CA	\$20.00	0.08	222	Bottle Opener				
290	11/30/09	1123	Williams	Mary	CA	\$50.00	0.08	555	Cheese Tray				
234	11/23/09	2242	DeBerry	Ann	DC	\$50.00	0.065	555	Cheese Tray				
237	11/7/09	2242	DeBerry	Ann	DC	\$50.00	0.065	111	Wine Guide				
238	11/10/09	2242	DeBerry	Ann	DC	\$40.00	0.065	444	Wine Gift Pack				
245	11/11/09	2242	DeBerry	Ann	DC	\$20.00	0.065	222	Bottle Opener				
250	11/18/09	2242	DeBerry	Ann	DC	\$20.00	0.065	222	Bottle Opener				
252	11/22/09	2242	DeBerry	Ann	DC	\$60.00	0.065	222	Bottle Opener				
253	11/23/09	2242	DeBerry	Ann	DC	\$60.00	0.065	222	Bottle Opener				
297	11/24/09	2242	DeBerry	Ann	DC	\$30.00	0.065	333	Jams & Jellies				
243	11/11/09	4254	Bailey	Ryan	MD	\$50.00	0.05	555	Cheese Tray				
246	11/18/09	4254	Bailey	Ryan	MD	\$30.00	0.05	333	Jams & Jellies				
248	11/22/09	4254	Bailey	Ryan	MD	\$60.00	0.05	222	Bottle Opener				
235	11/17/09	9500	Chin	April	KS	\$20.00	0.05	222	Bottle Opener				
242	11/23/09	9500	Chin	April	KS	\$30.00	0.05	333	Jams & Jellies				
244	11/24/09	9500	Chin	April	KS	\$20.00	0.05	222	Bottle Opener				
251	11/27/09	9500	Chin	April	KS	\$10.00	0.05	111	Wine Guide				

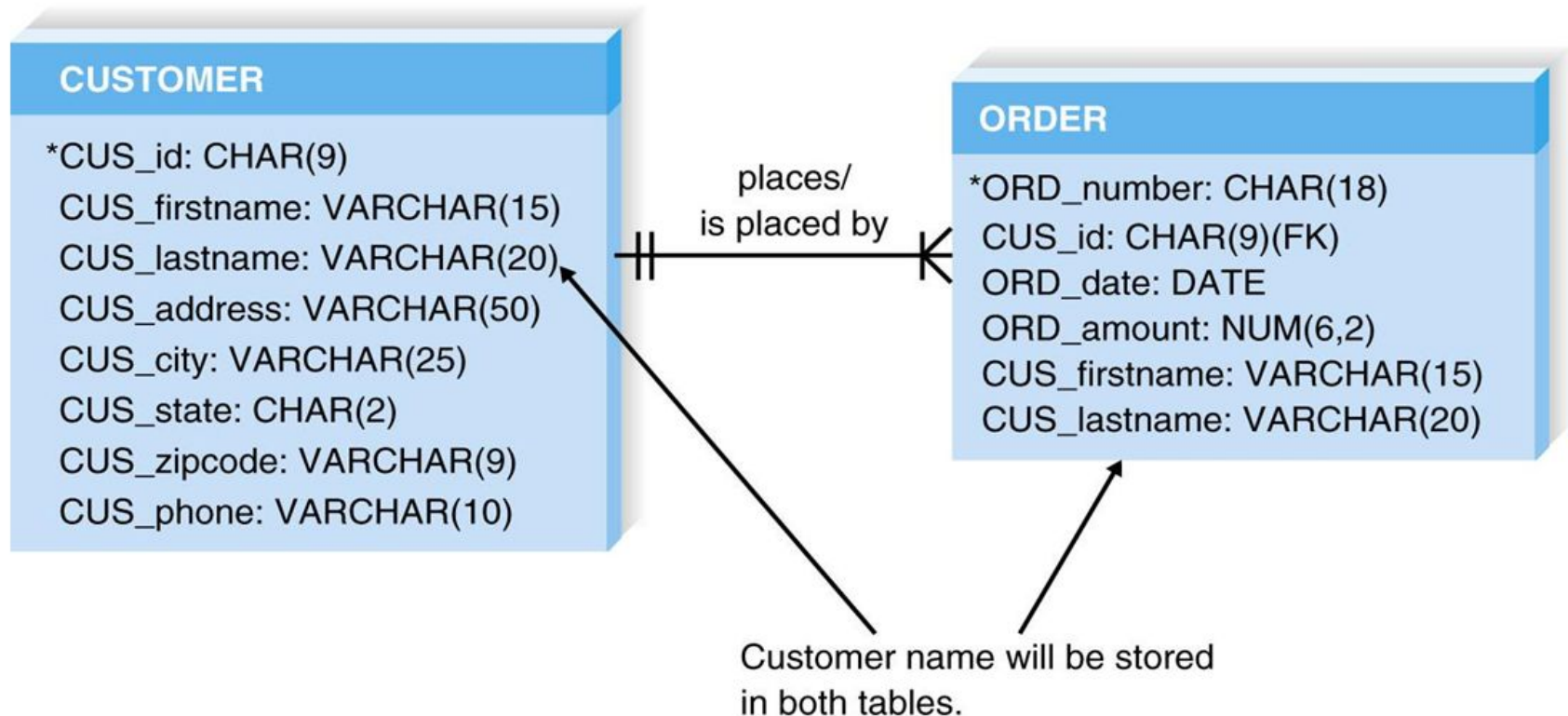
FIGURE 11-16
Optimizing Data Storage

Optimizing Access Speed

- After having optimized the data model design for data storage efficiency, the end result is that data are spread out across a number of tables.
- For a large relational database, it is necessary to optimize access speed.
- There are several techniques of optimizing access speed:
 - Denormalization
 - Clustering
 - Indexing
 - Estimating the size of data for hardware planning

Denormalization

- **Denormalization** – adding redundancy back into the design.



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There are four reasons for Denormalization.

Reason	Description	Example
Look-up Table	Include a code's description in the table using that code if the description is often used.	ORDER *ORD_number: CHAR(18) ORD_date: DATE ORD_amount: NUM(6,2) PAY_type: CHAR(2)(FK) PAY_description: VARCHAR(15) PAYMENT_TYPE *PAY_type: CHAR(2) PAY_description: VARCHAR(15)
1:1 Relationships	Combine tables if they are related 1:1 and if they usually are accessed together.	ORDER *ORD_number: CHAR(18) ORD_date: DATE ORD_amount: NUM(6,2) SHI_state: CHAR(2) SHI_method: CHAR(4) SHIPMENT *SHI_id: CHAR(9) ORD_num: CHAR(18) (FK) SHI_address: VARCHAR(50) SHI_city: VARCHAR(25) SHI_state: CHAR(2) SHI_zip: VARCHAR(9) SHI_method: CHAR(4)
1:N Relationships	Place fields from the parent (1) table into the child (N) table if the parent fields are used frequently with child information.	CUSTOMER *CUS_id: CHAR(9) CUS_firstname: VARCHAR(15) CUS_lastname: VARCHAR(20) CUS_address: VARCHAR(50) CUS_city: VARCHAR(25) CUS_state: CHAR(2) CUS_zipcode: VARCHAR(9) CUS_phone: VARCHAR(10) ORDER *ORD_number: CHAR(18) CUS_id: CHAR(9)(FK) ORD_date: DATE ORD_amount: NUM(6,2) CUS_firstname: VARCHAR(15) CUS_lastname: VARCHAR(20)
Star Schema Design	Data marts often are modeled with star schema design, which uses denormalization to maximize DSS query performance.	CUSTOMER *CUS_id: CHAR(9) CUS_firstname: VARCHAR(15) CUS_lastname: VARCHAR(20) CUS_address: VARCHAR(50) CUS_city: VARCHAR(25) CUS_state: CHAR(2) CUS_zipcode: VARCHAR(9) CUS_phone: VARCHAR(10) CUS_gender: CHAR(1) CUS_birthdate: DATE ORDER *ORD_number: CHAR(18) ORD_paytype: CHAR(2) ORD_shipstate: CHAR(2) ORD_shipmethod: VARCHAR(8) TIME *TIM_date: DATE TIM_dayofweek: NUM(1) TIM_weeknumber: NUM(2) TIM_monthnumber: NUM(2) TIM_quarter: NUM(1) TIM_fiscalyear: NUM(4) TIM_holidayflag: CHAR(1) FACT *FACT_id: CHAR(8) FACT_orderamount: NUM(6,2) FACT_ordercost: NUM(6,2) CUS_id: CHAR(9)(FK) TIM_date: DATE (FK) ORD_number: CHAR(18)

Clustering

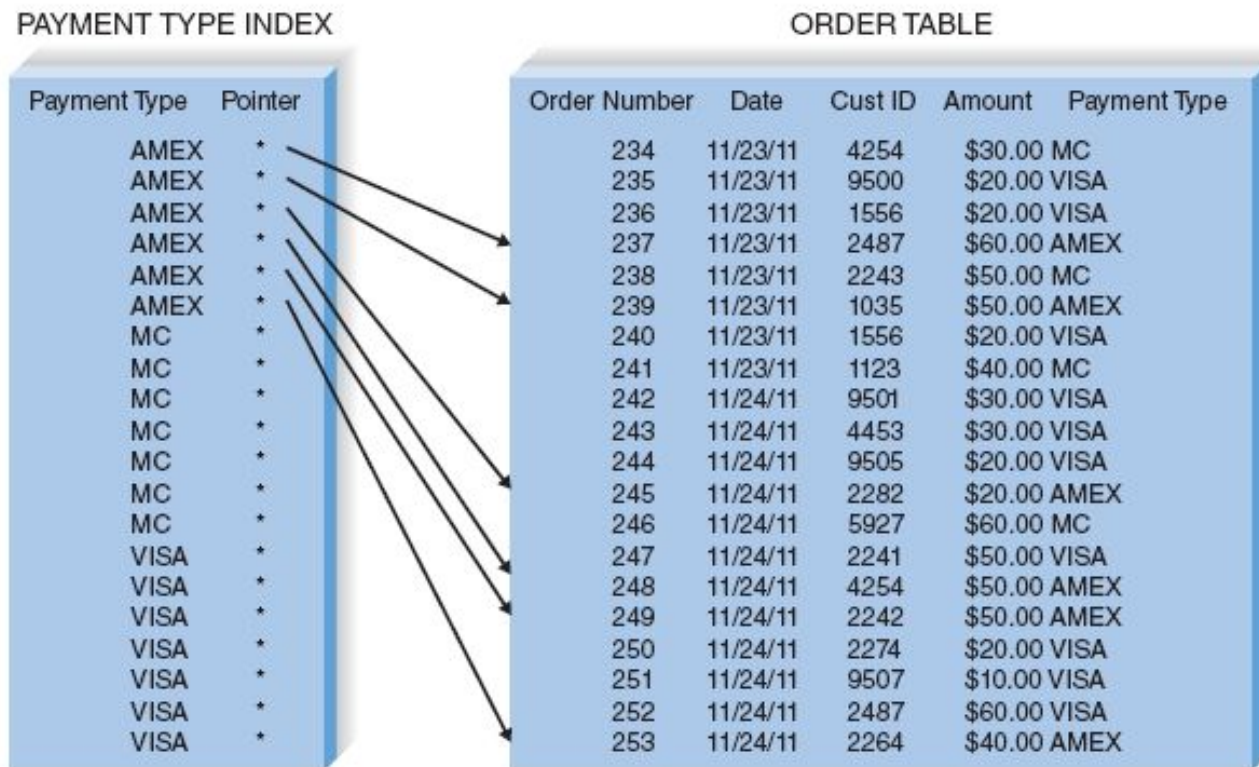
- **Clustering** – placing records together physically so that like records are stored close together.
- **Intrafile clustering** – Similar records in the table are stored together.
- **Interfile clustering** – Combining records from more than one table that typically are retrieved together.

Indexing

- An *index* in data storage is a minitable (similar to an index of a book) that contains values from one or more columns in a table and the location of the values within the table.
- Indexes require overhead in that they take up space on the storage.

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Example of indexing



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■ Guidelines for creating indexes

- Use indexes sparingly for transaction systems.
- Use many indexes to improve response times in decision support systems.
- For each table, create a unique index that is based on the primary key.
- For each table, create an index that is based on the foreign key to improve the performance of joins.
- Create an index for fields that are used frequently for grouping, sorting, or criteria.

Estimating Storage Size

■ ***Volumetrics*** – technique of estimating the amount of data that the hardware will need to support.

1. Calculate the amount of ***raw data*** - all the data that are stored within the tables of the database.
2. Calculate the ***overhead*** requirements based on the DBMS vendor recommendations.
3. Record the number of initial records that will be loaded into the table, as well as the expected growth per month.

Field

Average Size (Characters)

Order number	8
Date	7
Cust ID	4
Last name	13
First name	9
State	2
Amount	4
Tax rate	2
Record size	49
Overhead	30%
Total record size	63.7

Initial table size	50,000
Initial table volume	3,185,000

Growth rate/month	1000
Table volume @ 3 years	5,478,200

SUMMARY

■ File data storage formats

- **Files** are electronic lists of data.
- Five types of files: master, look-up, transaction, audit, and history.

■ Database storage formats

- A **database** is a collection of groupings of information
- A **DBMS** is software that creates and manipulates these databases.

■ Selecting a data storage format

- **Relational databases** support simple data types very effectively, whereas **object databases** are best for complex data.

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■ Physical entity relationship diagrams

- **Physical ERDs** contain references to how data will be stored in a file or database table, and **metadata** are included.

■ Optimizing data storage

- There are two primary dimensions in which to **optimize** a relational database: for **storage efficiency** and **for speed of access**.
- There are a number of techniques of optimizing data storage.